Qiwei Li

liqiwei152@gmail.com | 631-780-3888 | linkedin.com/in/qiwei-li-0464193a9 | github.com/Peoplerepabulic

Education

Washington University in St. Louis, McKelvey School of Engineering Aug. 2025 – May 2027
B.S. in Computer Science + Mathematics. Cumulative GPA: 3.75/4.00

Relevant Coursework: Matrix Algebra, Artificial Intelligence, Probability and Statistics, Data Science with Python

Manhattan University, Kakos School of Arts and Sciences Sept. 2023 – May 2025
Coursework toward B.S. in Computer Science. Dean's Honor List (all semesters)

Relevant Coursework: Discrete Structures, C++, Database Systems (SQL), Data Structures and Algorithms I & II

Experience

Back-End AI Engineering Intern, *FlyRank Corp.* Jul. 2026 – Aug. 2026
- Built an LLM usage-metering and billing API in FastAPI with idempotency-key guarantees: retried requests replay stored results, and a `UNIQUE` constraint prevents double-charging under concurrent retries; verified with 13 automated tests and 5 acceptance probes.
- Shipped an LLM classification endpoint with a provider abstraction, timeout and backoff retries, output schema repair, per-request cost logging, and a kill switch; scored 90% on a 10-case live-provider eval.
- Built an async job service (202 Accepted, status polling, idempotent replay, cron scheduling) and a report pipeline rendering multi-page PDFs from 200 seeded orders via headless Chromium.
- Developed a polite web scraper with robots.txt compliance, rate limiting, and schema validation (60/60 valid records).

Independent Researcher, *Manhattan University (Advisor: Prof. Alexander Adesman)* May 2024 – Oct. 2024
- Performed sentiment analysis on 2020 U.S. Election tweets using Python, Tweepy, TextBlob, and VADER, after filtering bot accounts by posting frequency, hashtag patterns, and account behavior.
- Visualized sentiment polarity by state with Matplotlib and Seaborn; analyzed how sentiment shifts tracked political events.

Independent Researcher, *Manhattan University (Advisor: Prof. Wafa Elmannai)* Jan. 2024 – May 2024
- Analyzed cloud computing adoption among SMEs, identifying data security and unpredictable costs as key barriers, supported by survey data showing 98% of cloud-using companies experienced breaches in 2020–2022.
- Designed a C++ simulation modeling SME annual profit under improved security and reduced redundant tech adoption.

Projects

Eedi: Mining Misconceptions in Mathematics (Kaggle) | *Python, Qwen2.5, LoRA* Sept. 2024 – Oct. 2024
- Earned a **Silver Medal (rank 42 / 1,446 teams, top 3%)** with an NLP pipeline matching math multiple-choice distractors to underlying student misconceptions, evaluated by MAP@25.
- Fine-tuned Qwen2.5-14B and Qwen2.5-32B-Instruct with contrastive learning, 4-bit quantization, and LoRA adapters; augmented training data with GPT-4o-generated synthetic examples.
- Ensembled embeddings from three LLMs for candidate retrieval, then reranked the top 25 with Qwen2.5-32B-Instruct-AWQ and bge-large-en-v1.5.

Settle – AI Group-Decision Agent for iMessage | *TypeScript, Node.js, LLM APIs* HackWashU 2026
- Built an AI agent that runs inside iMessage group chats and turns open-ended dining discussions into a final decision, using Photon Spectrum's real-time messaging SDK.
- Designed a decision state machine (collect → vote → tie runoff → veto → decide) that handles ties, vetoes, late joiners, and inactivity timeouts.
- Used LLM JSON-mode extraction with keyword-based fallbacks and persisted per-group preferences across sessions.

Skills

Languages: Python, C++, SQL, TypeScript
Frameworks & Libraries: FastAPI, React Flow, Inngest, pandas, scikit-learn, Matplotlib, Seaborn, TextBlob, VADER
ML / AI: NLP, LLM fine-tuning, LoRA, contrastive learning, embedding retrieval & reranking, AI agents, MCP
Tools & Platforms: Git, Docker, SQLite, REST APIs, OpenAI & Anthropic APIs, web scraping, async job processing

Leadership

Treasurer and Board Member, *Google Developer Groups of Manhattan University* Sept. 2023 – May 2025
- Managed monthly and quarterly financial statements and reporting; attended weekly board meetings.